Pharmacology of Antimuscarinic Agents (Muscarinic Antagonists)

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I am available to groups and individuals for pharmacology help and discussions, including Practice Questions, by appointment.

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Following this session, students will be able to:

1. **Structure-activity relationship → Pharmacokinetic properties**

Relate the physicochemical properties of the antimuscarinic agents with their pharmacokinetic properties (route of administration, absorption, distribution, metabolism, and excretion).

2. **Mechanism of action → Widespread Effects**

Correlate mechanism of action of atropine and “atropine-like” drugs with their effects on the organ systems (CNS, eye, heart, vessels, bronchi, gut, genitourinary tract, exocrine glands).

3. **Effects → Therapeutic uses**

Explain the clinical rationales for antimuscarinic use for each of the following therapeutic appplications: to a) prevent motion sickness; b) reduce respiratory secretions; b) produce mydriasis and cycloplegia; c) treat parkinsonism; d) relieve intestinal, biliary, and urinary spasm; and e) reverse the muscarinic toxicity of insecticides and nerve agents used in warfare. Give specific examples from the list of starred (✳) drugs.

4. **Mechanisms of action → Adverse effects → Contraindications**

Relate the mechanisms of the anticholinergic adverse effects with the contraindications for each of the starred (✳) drugs (drug-disease interactions).

5. **Mechanism of action → Effects of high dose → Toxicity → Specific antidote**

Explain the clinical rationale for the use of the specific antidote for anticholingeric overdose by relating its actions and effects to the signs and symptoms of the anticholinergic toxicity (anticholinergic toxidrome).

Preparation Materials

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Required

- **ScholarRx Bricks** | **Practice Questions** (not graded)

Supplemental resources if you wish to use them (they are not required:

- Video lecture | Dr. Goldstein's Word handout | Guided Reading Questions
- Links to a textbook and pharmacology review books are provided on the next slide.

STUDY TIPS:

- I provide several learning options with links so your are aware of them.
- ***You do not need to study all of them. Use the resources that <u>you</u> like to use for learning.***
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- Practice questions are intended to help you integrate and apply pharmacology concepts in the context of case vignettes and will help you become familiar with board-style of exam questions.

Links to resources

ScholarRx Bricks:

General Pharmacology > Autonomic Drugs > Drugs that Alter Parasympathetic Tone > Parasympatholytics > Antimuscarinic Drugs

Link: https://exchange.scholarrx.com/brick/muscarinic-antagonists

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2023

Chapter 8: Cholinoceptor-Blocking Drugs & Cholinesterase Regenerators

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281747427

REVIEW BOOKS HAVE PRACTICE QUESTIONS

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology; 8e, 2023; Chapter 5: Cholinergic Antagonists https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com/content.aspx?sectionid=253325538&bookid=3222

Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 8: Cholinoceptor-Blocking Drugs & Cholinesterase Regenerators

https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com/content.aspx?bookid=3461&sectionid=285590012

The following key points are intended help the student grasp the main ideas and concepts by organizing and summarizing the "need to know" information.

Key points may also serve as a concise reference to facilitate review.

Key points: Structure-activity relationship → Pharmacokinetic properties

- Antimuscarinic agents – muscarinic receptor blocking drugs – resemble acetylcholine structurally. An intact tropic acid ester and free -OH group in the acyl portion of the ester are essential for activity.
- The naturally occurring alkaloids, atropine and scopolamine, are tertiary amines, which are readily absorbed from the GI tract and conjunctival membranes, widely distributed including the CNS, and metabolized by the liver.
- Atropine is administered parenterally; unchanged drug and metabolites are excreted in the urine.
- Scopolamine is available in the US only as a transdermal patch that delivers drug to the systemic circulation for 72 hours. Scopolamine is rapidly and fully distributed to the CNS where it has greater effects than most other anticholinergic drugs. Synthetic versions are for oral use.
- The quaternary amines are less lipid soluble, charged synthetic compounds. They have reduced absorption from the GI tract, more peripheral effects, and are relatively free of CNS effects at low doses.

Key points: Mechanism of action → Widespread Effects

- The drugs in this class are competitive antagonists or inverse agonists of acetylcholine at muscarinic receptors.
  - Evidence indicates that muscarinic receptors are constitutively active and most drugs that block acetylcholine's actions are inverse agonists that shift the equilibrium to the inactive state of the receptor.
- Muscarinic receptors are extensively distributed in the periphery (heart, smooth muscle, exocrine glands, eyes) and the CNS. mAChRs $M_{1,3,5}$ are GqPCRs that ↑IP3, ↑DAG and cause smooth muscle contraction and stimulate secretions from exocrine gland. mAChRs $M_{2,4}$ are inhibitory Gi/oPCRs that ↓cAMP, ↑K+ channel current.
- Atropine and scopolamine are nonselective. They have affinity for all 5 subtypes of muscarinic receptors, $M_{1\text{-}5}$. Therefore, the drug binds to mAChRs in the intended tissue for the therapeutic effect and to mAChRs in "unintended" tissue → side effects.
- Some synthetic derivatives developed for relief of bladder spasm are somewhat selective for M_3 receptors and are associated with anticholinergic side effects, such as dry mouth, blurred vision, constipation, difficulty urinating, and dizziness.

Key points: Effects → Therapeutic uses

- Antimuscarinic agents have been used in the treatment of a wide variety of clinical conditions to inhibit parasympathetic activity in the respiratory tract, urinary tract, GI tract, eye, and heart.
- Their CNS applications are the prevention of motion sickness and the management of parkinsonian (extrapyramidal) symptoms.
- Atropine’s uses are:
  - preanesthesia medication to reduce salivation and excessive respiratory tract secretions
  - short-term intervention to prevent persistent bradycardia due to high vagal tone, drug-induced effect, and high-degree AV block
  - prior to reversal of neuromuscular blockade by cholinesterase inhibitors to decrease muscarinic effects
  - reversal agent (antidote) for cholinesterase poisoning by blocking the muscarinic effects in the periphery and CNS. For organophosphate poisoning, atropine is used with a cholinesterase regenerator.
- Scopolamine transdermal is for prevention of motion sickness, prevention of postop nausea/vomiting, and sialorrhea associated with neuromuscular disease.
- Inhaled antimuscarinics are for management of COPD and asthma; ophthalmics induce mydriasis and cycloplegia; others are for relief of GI and urinary spasms.

Key points: Mechanisms of action → Adverse effects → Contraindications

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- Because muscarinic receptors are extensively expressed, the drugs are associated with numerous adverse effects: dizziness, somnolence, impaired cognition; blurred vision due to mydriasis and cycloplegia; dry mouth, dyspepsia, constipation; tachycardia; urinary hesitancy, urinary retention; dry skin and suppression of thermoregulation by inhibiting sweating
- Antimuscarinic drugs should be avoided in patients with narrow-angle glaucoma, tachycardia, GI obstruction, paralytic ileus, severe ulcerative colitis, and prostatic hyperplasia.
- Antimuscarinic agents can:
  - increase intraocular pressure and aggravate narrow-angle glaucoma
  - exacerbate tachycardia by blocking vagal action on the sinoatrial node
  - aggravate GI obstruction, paralytic ileus, and ulcerative colitis by inhibiting GI motility
  - precipitate severe urinary retention in patients with prostatic hyperplasia
- Infants and young children are especially sensitive to the hyperthermia effects of antimuscarinic drugs.
- In older individuals, antimuscarinic agents may exacerbate or precipitate impaired cognition, confusion and psychosis associated with dementia, constipation, urinary retention, cause dizziness and drowsiness, and lead to falls.

Mechanism of action → Effects of high dose → Toxicity → Specific antidote

- The signs and symptoms of antimuscarinic overdose are summed up in the anticholinergic toxidrome – red as a beet, dry as a bone, blind as a bat, hot as Hades, mad as a hatter, the bowel and bladder lose their tone, and the heart runs alone.
- Toxic doses of atropine and scopolamine cause agitation, delirium, hallucinations, paranoia, seizures, coma.
- Quaternary ammonium muscarinic receptor antagonists can cause skeletal muscle weakness in overdose by inhibiting nicotinic receptors in the skeletal muscle junction.
- Physostigmine is the reversal agent for severe life-threatening central anticholinergic toxicity, if not contraindicated.
- Physostigmine must administered carefully. It can cause cholinergic toxicity with bradyarrhythmia, asystole, and seizures.
- Note: First-generation antihistamines, certain antipsychotic agents, tricyclic antidepressants, and others have antimuscarinic effects. Overdoses of these drugs produce syndromes that include features of atropine intoxication. Physostigmine should not be administered.

Muscarinic Receptor Antagonists

Atropa Belladonna

Datura stramonium
(Jimson weed)

Drugs to know by name along with their specific pharmacology features – in addition to the class effects, of course.

Tertiary Amines (cross BBB)	Quaterary Ammoniums (do not cross BBB)
✳Atropine	✳Ipratropium (SAMA)
✳Scopolamine	✳ Tiotropium (LAMA)
✳L-Hyoscyamine	✳Glycopyrrolate
✳Benztropine	
✳Oxybutynin	

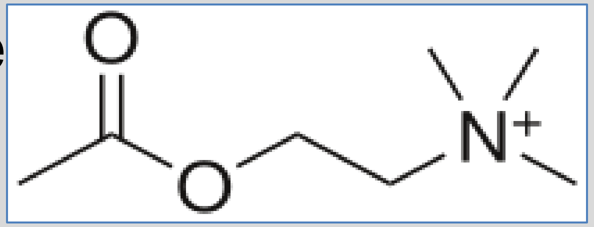
BBB: blood-brain barrier

SAMA: Short-acting muscarinic antagonist

LAMA: Long-acting muscarinic antagonist

Many other antimuscarinic drugs have been synthesized.
Note: L-Hyoscyamine is the levorotary stereoisomer of atropine for oral use.

3D structure resembles acetylcholine

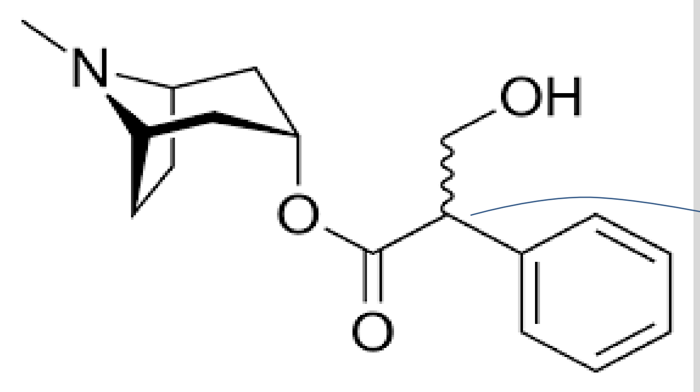


Acetylcholine

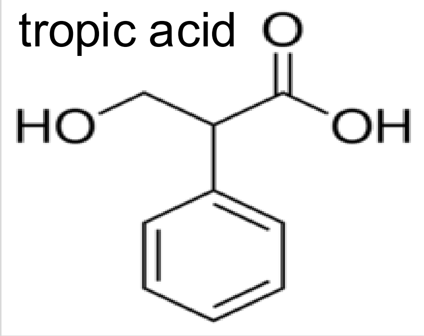
Belladonna alkaloids (Tropane alkaloids)

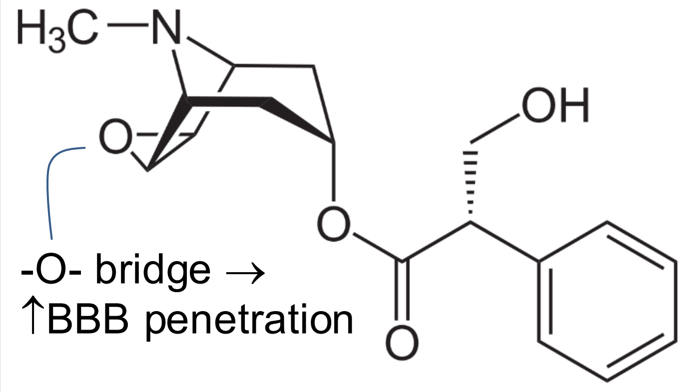
- **Tertiary amine esters**
  - Atropine (*d,l*-hyoscyamine)
  - Scopolamine (*d,l*-hyoscine)
  - *l*(-)-Hyoscyamine
  - and synthetic derivatives

- **Quaternary ammonium antagonists** (synthetic)
  - Iprotropium
  - Glycopyrrolate
  - Methscopolamine
  - Several others

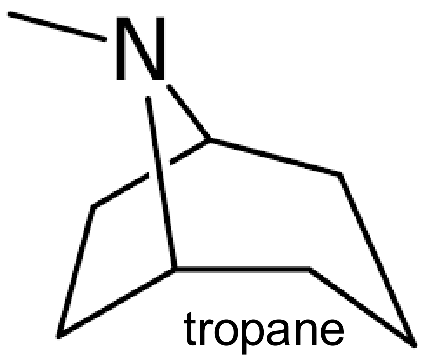


Atropine
tertiary amine
asymmetric C atom



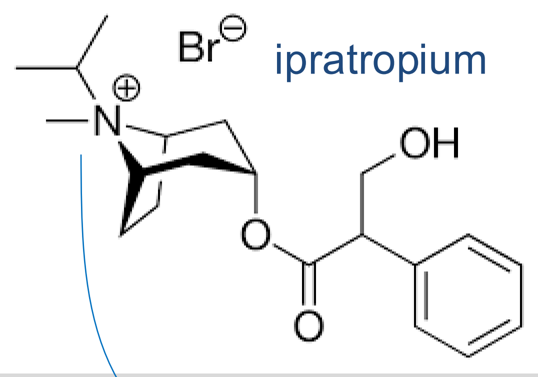


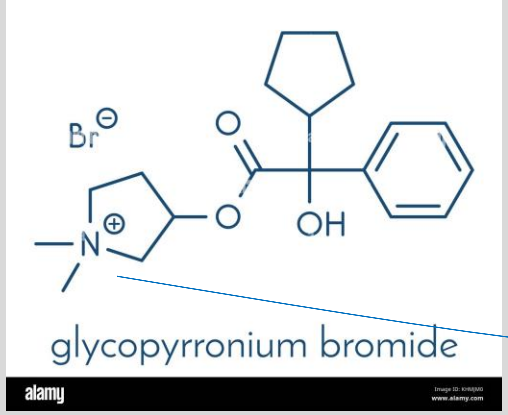
Scopolamine
tertiary amine



Structure-activity Relationship:

Tropic acid ester and free -OH group are essential for activity.





Ipratropium and Glycopyrrolate
quaternary ammonium

Pharmacokinetics

	Tertiary Amines		Quaternary amines
	Atropine	Scopolamine	Glycopyrrolate / Ipratropium
A	Nonpolar: Rapidly and well absorbed from GI tract and mucosal surfaces (all dosage forms)		Polar: Poorly and unreliably absorbed
	IV, IM, subQ, ophthalmic drops	Transdermal patch is the only form available in US	oral, inhalation, injection (product specific routes)
D	Widely distributed		Poor penetration of BBB · Minimal CNS effects at low therapeutic doses
	Dose-dependent CNS levels	Scopolamine is rapidly and fully distributed into the CNS at therapeutic doses.	
M	Hepatic, enzymatic hydrolysis Not available in oral formulations.	Extensive Hepatic first-pass effect Not available orally for systemic effect.	Product specific Low bioavailability
E	Urine, unchanged drug and metabolites	Minimal excretion in urine (scopolamine base in transdermal patch)	Urine, unchanged mainly Feces: unabsorbed drug
t½	Atropine 2-3 hours Increased in children <2 yo and older adults >65 yo	Transdermal Scopolamine base: 9.5 h (IM, IV 1-4 h for comparison) Duration (patch): 72 hours	Product specific

Scopolamine is available in Canada and other countries in oral and injectable forms for additional indications.
BBB: blood brain barrier

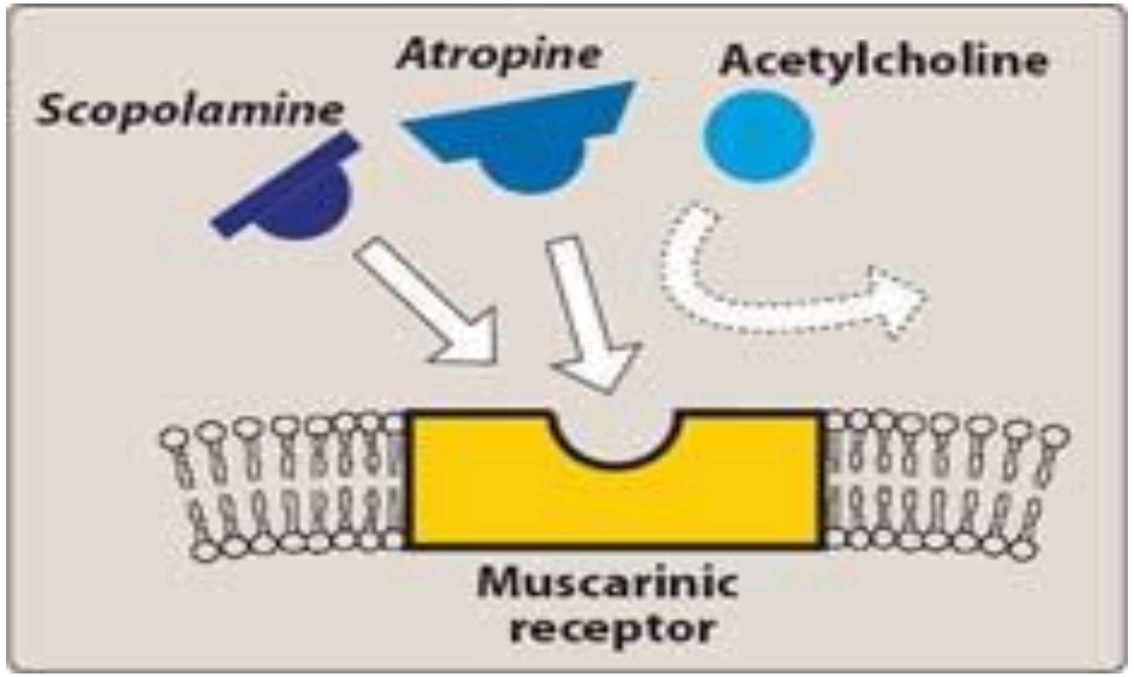
Mechanism of action

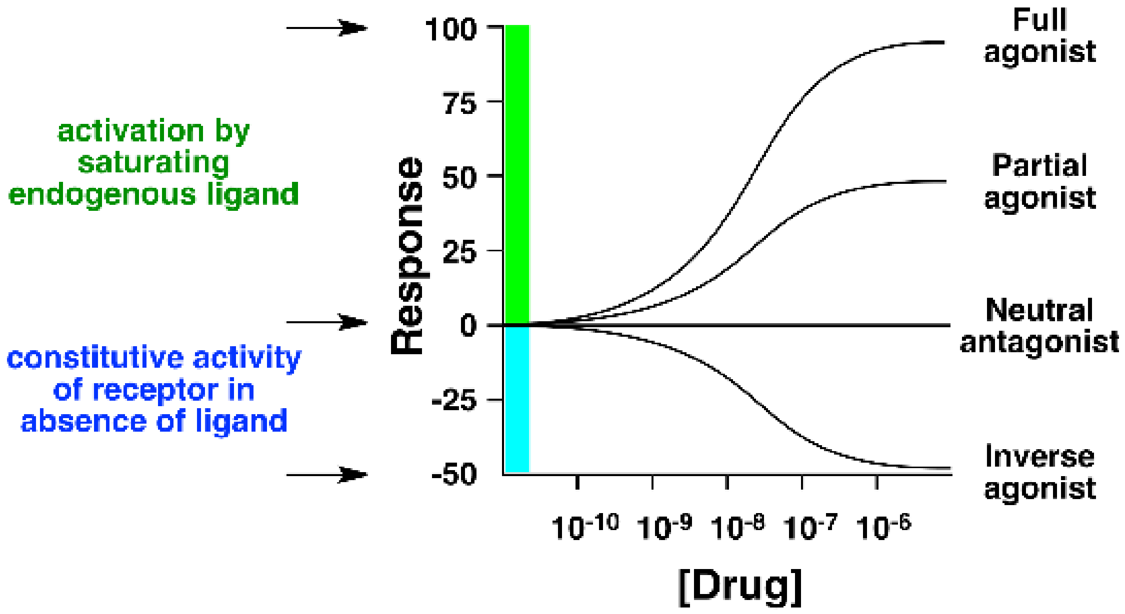
- ***Competitive antagonists at muscarinic receptors*** ⇒ prevent ACh from binding to its receptors the periphery and the CNS

- ***Inverse agonists***

Evidence indicates that muscarinic receptors are constitutively active, and most drugs that block the actions of acetylcholine are ***inverse agonists*** that shift the equilibrium to the inactive state of the receptor***.***

These include atropine, ipratropium, glycopyrrolate, pirenzepine, trihexyphenidyl, dicyclomine, and methscopolamine.





Upper figure: Lippincott Illustrated Reviews: Pharmacology Figure 5.3
Lower figure: Wikipedia By Boghog - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=34657034

Responses of Effector Organs to Muscarinic Antagonism

Salivary, Lacrimal glands	**Decrease salivation and tearing → dry mouth and dry eyes**
Sweat glands	Decrease sweating → hot, dry skin → **hyperthermia** may result
Eyes	Block miosis and accommodation → mydriasis and cycloplegia → **blurred vision**
Pulmonary	Direct effects: ↓ respiratory secretions Indirect effects: Block vagal reflex bronchoconstriction by histamine, bradykinin, eicosanoids → Allows unopposed beta-adrenergic **bronchodilation** – a desirable effect
Cardiovascular	Inhibit vagal-mediated effects on M_2 receptors: in SA nodal pacemaker → **tachycardia** | in AV node → **enhance conduction in the AV node** Note: **Paradoxical bradycardia** (see slide 19)
GI	Inhibit smooth muscle contractions→ inhibits motility→ **Delayed gastric emptying / Constipation** Inhibits gastric acid secretion
Urinary	Relax smooth muscle of bladder walls (detrusor muscle) → **slows voiding** Inhibits urethral smooth muscle relaxation → **urinary retention**
CNS	Euphoria, drowsiness, anterograde amnesia, fatigue, dreamless sleep

Atropine's Dose-Related Effects

DOSE	EFFECTS
0.5 mg	Slight cardiac slowing; some dryness of mouth; inhibition of sweating
1 mg	Definite dryness of mouth; thirst; acceleration of heart, sometimes preceded by slowing; mild dilation of pupils
2 mg	Rapid heart rate; palpitation; marked dryness of mouth; dilated pupils; some blurring of near vision
5 mg	Above symptoms marked; difficulty in speaking and swallowing; restlessness and fatigue; headache; dry, hot skin; difficulty in micturition; reduced intestinal peristalsis
≥10 mg	Above symptoms more marked; pulse rapid and weak; iris practically obliterated; vision very blurred; skin flushed ("atropine flush"), hot, dry, and scarlet; ataxia, restlessness, and excitement; hallucinations and delirium; seizures; coma
	Please see slide 29 for anticholinergic toxidrome mnemonic

CNS Sx

Marked CNS Sx

Delirium Coma

Paradoxical Bradycardia

A transient effect of low-dose I.V. atropine

Normally, ACh activation of PREjunctional muscarinic autoreceptors may suppress ACh release

Low-dose atropine may block this inhibitory action

Higher atropine doses block muscarinic M2 effects on SA node
↓
parasympatholytic effects
↓
tachycardia

May cause release of ACh, which slows heart rate

Paradoxical bradycardia can be avoided with rapid intravenous injection of atropine.

http://www.atlasophthalmology.com/atlas/photo.jsf?node=5820&locale=en

Left eye – pharmacologic mydriasis:

- No response to light and near reflex in the left eye
- The left pupil remains dilated after instillation of 1% pilocarpine in the left eye.

Therapeutic Uses of Selected Antimuscarinic Agents

Atropine	• Pre-anesthesia to ↓ salivary and respiratory tract secretions; to prevent bradycardia by some anesthetics • Prior to administration of neostigmine or pyridostigmine (cholinesterase inhibitor) for reversal of neuromuscular blockade to reduce muscarinic side effects • Cardiovascular: Short-term intervention for persistent bradyarrhythmia due to high vagal tone, drug-induced, high-degree AV block • Reversal agent: Cholinergic poisoning to block ACh actions in CNS and periphery
Scopolamine	Available only in transdermal patch in the U.S. • Motion sickness prevention • Prevention of post-op nausea/vomiting • Control of excessive salivation in neuromuscular diseases
Ipratropium (SAMA) Tiotropium (LAMA) Others	Oral inhalation → local action on bronchial tissues • Block parasympathetic constriction of bronchial smooth muscle and reduce secretions • Management of COPD (chronic bronchitis and emphysema) and asthma (SAMA for acute relief; LAMA preferred for maintenance) • Nasal spray for relief of runny nose due to colds
Glycopyrrolate	• Induction of anesthesia and intubation and intraoperative use to block vagally-mediated bradycardia and to reduce salivary, tracheobronchial, and GI secretions • Prior to administration of neostigmine or pyridostigmine to reduce muscarinic side effects • Reduction of secretions at end of life (death rattle) • Hyperhidrosis, primary axillary (topical cloth towelette or oral)

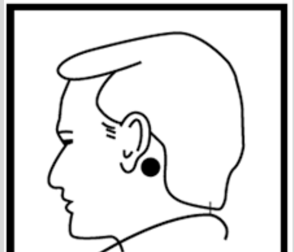


Figure 3

Therapeutic Uses of Selected Antimuscarinic Agents

Oral products

Oxybutynin Others	• Antispasmodic for symptomatic treatment of overactive bladder • Moderately selective for M_3 receptors • Extensive hepatic metabolism (CYP3A4) • High incidence of antimuscarinic side effects
L-Hyoscyamine	• Relief of visceral spasms (other drugs: dicyclomine and Donnatal = belladonna alkaloids with phenobarbital)
Benztropine Trihexyphenidyl	• Parkinson's disease: manage tremors due to relative excess of acetylcholine • Treatment of drug-induced extrapyramidal symptoms (parkinsonism)

Ophthalmic products

Atropine Homatropine Cyclopentolate Tropicamide	Ophthalmic drops to induce mydriasis and cycloplegia Duration of action: Atropine ~3 days | Homatropine 12 hours Cyclopentolate 24 hours | Tropicamide ~6 hours

Summary of Therapeutic Uses by Organ System

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Organ	Comment	Drugs
CNS	Motion sickness prevention Preanesthesia sedative, anterograde amnesia	Scopolamine
	Parkinson disease, EPS: adjunctive agents	Benztropine / Trihexyphenidyl
Cardiac	Counteract reflex bradycardia	Atropine
Respiratory tract	Preanesthesia: block secretions	Atropine Glycopyrrolate
	Inhaled bronchodilators for management of COPD and asthma	Ipratropium (short-acting), Tiotropium (long-acting), several other long-acting drugs
GI tract	Antispasmodic	L-Hyoscyamine / Dicyclomine Donnatal (belladonna / phenobarbital)
Urinary bladder	Overactive bladder antispasmodic	Oxybutynin Several others
Eyes	Mydriasis and cycloplegia for examination of optic disc and retina	Tropicamide Cyclopentolate
Reversal agent	Carbamate / organophosphate cholinesterase inhibitor toxicity Blocks muscarinic receptors in CNS and periphery (used with pralidoxime for organophosphate poisoning)	ATROPINE

Adverse Effects Resulting from Muscarinic Blockade

(not a complete list)

- Muscarinic blockade causes:
  - CNS effects
    - Dizziness
      Cognitive impairment
      Delirium
    - Scopolamine causes CNS effects at therapeutic doses
  - Xerostomia
    Blurred vision
  - Thick respiratory secretions, plugging
    - Systemic drugs
    - Inhaled antimuscarinics do not cause it.
  - Dyspepsia
    Constipation
    Urinary retention
  - Hot, dry skin
    by suppression of thermoregulation
    - FDA Warning: Transderm Scopolamine
      Hyperthermia resulting in hospitalization has occurred in children <17 and adults ≥60 years old.

Contraindications* for Use of Atropine-Like Agents

***Atropine is not contraindicated in cases of organophosphate toxicity.**

Narrow-angle glaucoma	Pupil dilation (mydriasis) may block the filtration angle, which may block outflow of aqueous humor through trabecular meshwork and canal of Schlemm, leading to increased intraocular pressure
Prostatic hyperplasia	Aggravate or precipitate urinary retention
Cardiovascular disease	Tachycardia can precipitate ischemic attack, myocardial infarction, or arrhythmia
Gastrointestinal obstruction Paralytic ileus Severe ulcerative colitis	Inhibition of GI motility can aggravate these conditions.

Toxicity: Antimuscarinic Drugs

TOXIC SYNDROME

- pulse rapid and weak
- iris practically obliterated; vision very blurred
- skin flushed, hot, dry, and scarlet
- ataxia, restlessness, and excitement
- hallucinations and delirium
- seizures, coma

The anticholinergic toxidrome mnemonic is on the next slides.

REVERSAL: Physostigmine

- Respiratory, CV support, seizure treatment (benzodiazepine)

Use of physostigmine is controversial.

- Most patients with anticholinergic toxicity do well on supportive therapy alone.
- ***Contraindication***: Overdose by tricyclic antidepressants (TCA), phenothiazine derivatives, 1st-gen antihistamines, and other drugs with "atropine-like" effects have a wide range of effects, including cardiac toxicity. Physostigmine can exacerbate them and lead to severe complications.

Physostigmine must be administered very cautiously.
Rapid administration can cause symptoms of cholinergic excess and cholinergic toxicity,

- bradyarrhythmia / asystole / seizures / respiratory depression

Mnemonic: Anticholinergic Toxidrome

Red as a beet	flushed (scarlet), hot skin (“atropine flush”)
Dry as a bone	dry mouth; dry skin; reduced GI secretions
Blind as a bat	large, nonreactive pupils; blurred vision
Hot as Hades	hyperthermia; tachycardia; hypertension (“atropine fever”)
Mad as a hatter	confusion, disorientation, hallucinations, seizures, coma
The bowel and bladder lose their tone	effects on smooth muscle and secretions → absent bowel sounds and urinary retention
And the heart runs alone	tachycardia; weak, rapid pulse

Red as a Beet
“atropine flush”
due to hyperthermia

Dry as a Bone
secretions inhibited

Blind as a Bat
mydriasis, cycloplegia

Hot as a Hades
hyperthermia; tachycardia; hypertension

Mad as a Hatter
In olden days, hatters breathed mercury fumes from the felt used in hat making.

Special Populations

Pregnancy: If there is a clear indication for use, medications used as antidotes (atropine) should not be withheld because of fears of teratogenicity.

Infants and young children: Especially susceptible to the ***hyperthermic effects*** and other toxic effects of muscarinic antagonists

Elderly: Susceptible to both peripheral and CNS effects, which may exacerbate or precipitate many conditions, including impaired cognition, confusion and psychosis associated with dementia, and lead to falls.

- ***Drugs with anticholinergic effects should be avoided in older individuals.*** The side effects of the drugs outweigh the benefits.

Check your knowledge

A hospitalized patient is diagnosed with symptomatic sinus bradycardia. What drug can be administered?

The drug is administered by rapid IV push to avoid a specific adverse effect. What is the adverse effect and the proposed mechanism?

Janny is a 38-year-old woman prone to carsickness. She places a small patch that looks like a round Band-Aid behind her ear the day before she and a companion are leaving for a road trip across the country. What is the drug? Other than convenience, what is an advantage of this route of administration?

Check your knowledge

An otherwise healthy 77-year-old female presents to the family practice clinic with signs and symptoms of a respiratory infection that would likely respond to a macrolide antibiotic. The patient takes oxybutynin for urinary incontinence due to bladder spasms. Which macrolide antibiotics could lead to an increase in the antimuscarinic side effects of oxybutynin? (Hint: PK mechanism)

A 62-year-old male smoker diagnosed with chronic obstructive pulmonary disease (COPD) is given prescriptions for a long-acting antimuscarinic drug (LAMA) and a short-acting beta agonist (SABA) and guidance on smoking cessation. What are the drugs and how will the patient take them?

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Check your knowledge

A 30-year-old female presents to the ED with confusion; delirium; hot, flushed, dry skin; dilated pupils; tachycardia; hypotension; conduction anomalies; and absent bowel sounds. She has a seizure while in the ED. Her roommate shows the health care team an empty prescription vial labeled amitriptylline 50 mg daily and dated yesterday. (Amitriptyline is a tricyclic antidepressant). Why should the reversal agent for antimuscarinic toxicity be avoided in this patient?

A 20-year-old pregnant female agricultural worker is brought to the emergency room with confusion and other signs and symptoms of cholinergic toxicity consistent with poisoning by insecticides. Her coworker reports that carbamate, not organophosphate, insecticides are used. She is treated with decontamination, respiratory, and cardiovascular support. What is the reversal agent and should it be administered to this patient?

Summary of Antimuscarinic Drugs

- Anticholinergic agents block the actions of acetylcholine at the receptors.
- Atropine and scopolamine are antimuscarinic tertiary amines that are readily absorbed and penetrate the CNS. Synthetic quaternary amines are poorly absorbed and have low CNS effects.
- Atropine and scopolamine nonspecifically block all types of muscarinic receptors (M1-M5). Some agents are somewhat selective.
- Antimuscarinic drugs have many physiologic effects on circulation, respiration, alertness, vision, GI and urinary function.
- Therapeutic uses include with induction of anesthesia and intubation to prevent bradycardia and inhibit salivation and respiratory secretions; motion sickness prophylaxis; COPD and asthma; GI spasms; overactive bladder; tremors of Parkinson’s disease; induction of mydriasis; and sialorrhea.
- Atropine is the reversal agent for cholinergic poisoning.
- Physostigmine is the reversal agent for atropine poisoning.

- Adverse effects reflect blockade of cholinergic effects: dry mucous membranes and dry skin, blurred vision, urinary retention and constipation, tachycardia, dizziness, somnolence, impaired cognition, and hyperthermia.
- Young children are particularly susceptible to hyperthermia.
- Elderly patients are sensitive to the peripheral and CNS adverse effects, which can aggravate constipation and urinary retention, cause confusion and psychosis, and dizziness and somnolence, and lead to falls.
- ➢ Drugs with anticholinergic effects should be avoided in the older patient.
- Anticholinergics are contraindicated in patients with narrow-angle glaucoma, prostatic hyperplasia, tachycardia, GI obstruction, paralytic ileus, or severe ulcerative colitis.

Check your knowledge

A hospitalized patient is diagnosed with symptomatic sinus bradycardia. What drug can be administered?

- Atropine blocks vagal effects that normally slow the heart rate, which allows unopposed sympathetic nervous system stimulation, resulting in increased heart rate and improved conduction through the AV node.

The drug is administered by rapid IV push to avoid a specific adverse effect of the drug. What is the adverse effect and the proposed mechanism?

- Paradoxical bradycardia. The initial low concentration of atropine from a slow intravenous injection may initially block inhibitory presynaptic receptors, causing ACh release, thus, slowing of the heart rate. Rapid IV administration avoids this effect.

Janny is a 38-year-old woman prone to carsickness. She places a small patch that looks like a round Band-Aid behind her ear the day before she and a companion are leaving for a road trip across the country. What is the drug? Other than convenience, what is an advantage of this route of administration?

- Trandermal scopolamine bypasses first-pass metabolism and maintains steady, consistent blood levels and avoids peaks and troughs. Avoiding high peak concentrations may reduce dizziness and drowsiness and other side effects.

Check your knowledge

An otherwise healthy 77-year-old female presents to the family practice clinic with signs and symptoms of a respiratory infection that would likely respond to a macrolide antibiotic. The patient takes oxybutynin for urinary incontinence due to bladder spasms. Which macrolide antibiotics could lead to an increase in the antimuscarinic side effects of oxybutynin?

- Erythromycin and clarithromycin are strong (potent) CYP3A4 inhibitors. Oxybutynin is metabolized by CYP3A4 and is associated with a constellation of antimuscarinic side effects.

A 62-year-old male smoker diagnosed with chronic obstructive pulmonary disease is given prescriptions for a long-acting antimuscarinic drug (LAMA) and a short-acting beta agonist (SABA) and guidance on smoking cessation. What are the drugs and how will the patient take them?

- The drugs are the LAMA, tiotropium or other long-acting drug for maintenance bronchodilation by blocking cholinergic bronchoconstriction, and the SABA, albuterol for direct stimulation of beta-2 receptors, which causes bronchial smooth muscle relaxation and bronchodilation. You will learn about SABAs and LABAs in the Sympathomimetics lectures.

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Check your knowledge

A 30-year-old female presents to the ED with confusion; delirium; hot, flushed, dry skin; dilated pupils; tachycardia; hypertension; conduction anomalies; and absent bowel sounds. She has a seizure while in the ED. Her roommate shows the health care team an empty prescription vial labeled amitriptylline 50 mg daily and dated yesterday. (Amitriptyline is a tricyclic antidepressant). Why should the reversal agent for antimuscarinic toxicity be avoided in this patient?

- Overdose by tricyclic antidepressants (TCA), phenothiazine derivatives, and other drugs with anticholinergic actions have a wide range of effects, including cardiac toxicity. Physostigmine can exacerbate them and lead to severe complications.

A 20-year-old pregnant female agricultural worker is brought to the emergency room with confusion and other signs and symptoms of cholinergic toxicity consistent with poisoning by insecticides. Her coworker reports that carbamate, not organophosphate, insecticides are used. She is treated with decontamination, respiratory, and cardiovascular support. What is the reversal agent and should it be administered to this patient?

- The reversal agent is atropine. Pralidoxime is not indicated because it is not effective for poisoning by carbamate insecticides. Yes, atropine should be administered because it is life-saving and the benefit far outweighs any risk to the fetus.

References

- Access Medicine Goodman and Gilman’s The Pharmacological Basis of Therapeutics 14e, 2022; Chapter 11: Muscarinic Receptor Agonists and Antagonists
- Access Medicine Katzung’s Basic & Clinical Pharmacology, 16e, 2024; Chapter 8: Cholinoceptor Blocking Drugs
- UpToDate Lexidrug: various drug monographs

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